

Exploratory Data Analysis

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Data-Characterization

The process of **describing** and **summarizing** the main characteristics of a **dataset** using **descriptive statistics** and visualization techniques to understand its structure, distribution, quality, and underlying patterns.

Types of Analytics



<https://medium.com/co-learning-lounge/types-of-data-analytics-descriptive-diagnostic-predictive-prescriptive-922654ce8f8f>

Types of Analytics

Descriptive Analytics – “What happened?”

- **Summarizes** historical data
- Uses statistics (**mean, median, variance**) and visualizations
- Identifies **patterns, trends**, and **distributions**
- Foundation of Exploratory Data Analysis (EDA)

Example: Monthly sales reports, summary dashboards.

Types of Analytics

Diagnostic Analytics – “Why did it happen?”

- Investigates **causes** behind observed outcomes
- Uses drill-down analysis, **correlations, comparisons**
- **Identifies relationships** and root causes

Example: Analyzing why sales dropped in a specific region.

Types of Analytics

Predictive Analytics – “What is likely to happen?”

- Uses **historical data** to **forecast** future outcomes
- Applies **statistical models** and machine learning
- Identifies probabilities and **future trends**

Example: Predicting customer churn.

Types of Analytics

Prescriptive Analytics – “What should we do?”

- **Recommends actions** based on predictions
- Uses optimization, simulation, and **decision** models
- Supports **strategic** and operational decision-making

Example: Recommending pricing strategies to maximize profit.

Data Meaning

Origin and Meaning of Data

- The term *data* comes from Latin and originally means “**something given.**”
- Traditionally, data has been described as **facts, observations, or information** provided as a **basis for reasoning, analysis, or decision-making.**
- In scientific and technical contexts, it may refer to **quantitative or qualitative values** obtained from observation, measurement, or experimentation.

Data in the Context of Data Science

- In Data Science, data refers to **structured or unstructured facts, measurements, or records** collected for analysis.
- These data are processed to extract insights, identify patterns, support decisions, and build predictive models.

Data Nature

Time Perspective

- Time-independent data: Not associated with a temporal dimension (e.g., static attributes like gender or blood type).
- Time-dependent data: Linked to time and often analyzed as sequences or trends (e.g., stock prices, sensor readings, logs).

Location Perspective

- Location-independent data: Data without geographic reference or spatial context.
- Location-dependent data: Data tied to physical or spatial locations, common in GIS, mobility, or environmental analysis.

Understanding temporal and spatial context helps reveal patterns, trends, and relationships in exploratory data analysis.

Data Nature

Primary Data

- Collected directly from original sources (experiments, surveys, sensors).
- Typically more specific but may require validation and cleaning.

Derived (Processed) Data

- Produced by transforming, aggregating, or restructuring existing data.
- Common in data pipelines, feature engineering, and reporting.

Theoretical or **Predicted** Data

- Generated from models, simulations, or forecasting techniques.
- Often used for scenario analysis, prediction, and decision support.

Recognizing data origin is essential for assessing reliability, bias, and analytical suitability.

Data Nature



Difference Between Primary & Secondary Data	
Primary Data	Secondary Data
• First-hand data • More reliable • Control over collection method • Time & money intensive	• Second-hand Data • Comparatively less reliable • No control over collection method • Time & cost effective

<https://unstop.com/blog/difference-between-primary-and-secondary-data>

Data Nature

Determinable Data (**deterministic**)

- Refers to **quantitative values** that can be assumed to have a specific and **well-defined value** under given conditions.
- **Variability is minimal or controlled**, meaning repeated measurements tend to produce the same result.
- Often associated with deterministic systems, where outcomes follow predictable rules.

Examples

- Physical constants or controlled laboratory measurements
- Engineered system outputs under fixed conditions
- Time-dependent data when time itself is explicitly controlled

These data are generally easier to model because uncertainty and randomness are limited.

Data Nature

Stochastic Data

- Refers to data where values naturally **fluctuate** due to **randomness**, variability, or measurement uncertainty.
- Even under similar conditions, **repeated observations may produce different results.**
- Common in real-world phenomena influenced by complex or unknown factors.

Examples

- **Environmental** and geoscience data (weather, seismic activity)
- Financial market data
- Human behavior and social data

Stochastic data typically require **probabilistic models**, statistical analysis, and uncertainty quantification in Data Science.

Data Type

Quantitative Data

- **Numerical values** measured in well-defined units
- Directly suitable for mathematical and statistical analysis
- *Example:* Temperature, income, time

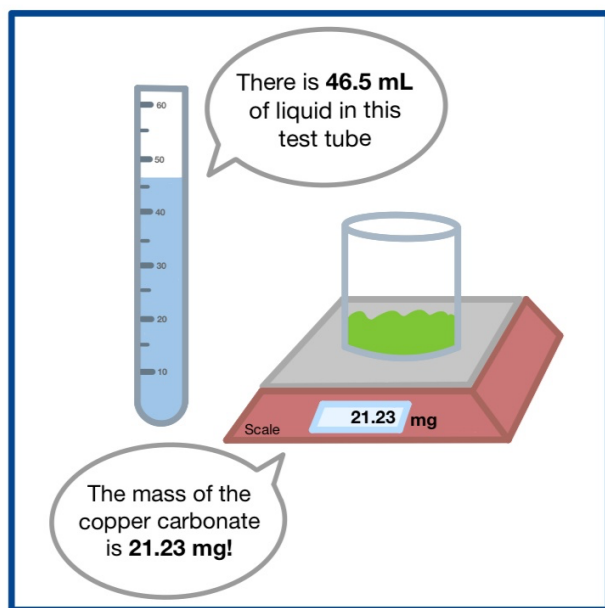
Qualitative Data

- **Descriptive or categorical information**
- Expresses attributes or characteristics rather than numbers
- *Example:* Interview responses, behavioral descriptions

Data Type

Quantitative Data

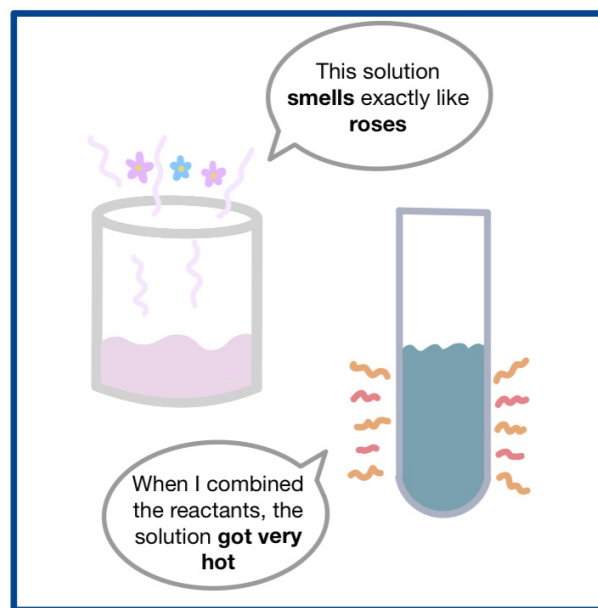
Based on numbers (measurement or counting)



VS

Qualitative Data

Based on observations (involves the senses)



Data Presentation

Numerical Data

- Presented as **numerical values**
- Typically derived from **quantitative measurements**
- *Example:* Temperature, sales figures, test scores

Graphical Data

- Represented **visually** through charts, graphs, maps, or models
- Helps simplify large datasets and support **visual interpretation**

Symbolic Data

- Expressed using **symbols**, codes, or specialized notation
- Common in fields like meteorology, science, or technical systems

Data Measure

Nominal Data

- Used to **label or categorize** items without implying order
- Numbers (if assigned) act only as identifiers, **not quantities**
- *Example:* Journal disciplines, gender categories, product types

Ordinal Data

- Represents categories with a meaningful **order or ranking**
- Differences between ranks are not necessarily equal
- *Example:* Satisfaction ratings (poor → excellent), education levels

Data Measure

Interval Data

- Ordered **numerical** data with equal intervals between values
- No true zero point, so ratios are not meaningful
- *Example:* Temperature in Celsius or Fahrenheit, calendar dates

Ratio Data

- Numerical data with equal intervals and a **true zero point**
- Allows meaningful comparisons using ratios (e.g., twice, half)
- *Example:* Weight, height, distance, income, price

Data Continuity

Continuous Data

- Can take **any value within a given range**
- Includes infinitely many possible values
- Typically obtained through measurement
- *Example:* Height, weight, temperature, time

Discrete Data

- Consists of **countable**, distinct values
- Usually obtained by counting rather than measuring
- Decimal values are typically not meaningful
- *Example:* Number of students, vehicles, or library members

Data - number of Characteristics

Univariate Data

- Observations based on a **single variable** or characteristic
- Focuses on describing one aspect of the data
- *Example:* Students' exam scores in one subject

Bivariate Data

- Involves **two variables** measured simultaneously
- Used to explore relationships or associations
- *Example:* Height and weight of students

Multivariate Data

- Includes three or **more variables** observed together
- Enables analysis of complex relationships and patterns
- *Example:* Income, education level, and savings data

Data - Reference of time

Time Series Data

- Collected sequentially over time
- Used to analyze trends, patterns, or temporal changes
- *Example:* Monthly sales, annual student enrollment, production over years

Cross-Sectional Data

- Collected at a single point in time across different units or groups
- Useful for comparisons between populations, regions, or categories
- *Example:* Income levels across regions in a given year

Data Structure

Structured Data

- Typically quantitative
- Organized in tabular formats (rows and columns)
- Easy to store, query, and analyze
- *Example:* Databases, spreadsheets

Unstructured Data

- Typically qualitative
- No predefined format or organization
- Often includes text, audio, images, or video
- *Example:* Emails, social media posts, multimedia files

Semi-Structured Data

- Combines elements of structured and unstructured data
- Uses flexible formats with some organization
- *Example:* JSON, XML, log files

Obrigado!